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Original/Overview Paper

Enhancing Music Classification via Source Separation and Feature Fusion

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ABSTRACT

Music classification tasks have been a central focus within Music Information Retrieval (MIR) for decades. Previous work usually applies a classifier to mixture audio for classification. However, some music attributes are usually associated with the presence or performance of certain instruments. Motivated by such an observation, this work explores whether the use of state-of-the-art music source separation (MSS) models benefits downstream music classification tasks. We first apply MSS models to separate the music signal into stems, including vocals, drums, bass, piano, and guitar. Then, we propose several fusion strategies that combine the features extracted from individual stems and the original mixture for music classification. Experiment results suggest that the use of source-separated audio stems improves the music classification performance for genre classification but is less effective for emotion classification, which is more related to the holistic semantics of the mixture audio. To our knowledge, this is the first work that studies the effectiveness of using MSS for different music classification tasks.

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1 Introduction

Music classification encompasses a variety of core tasks within Music Information Retrieval (MIR), ranging from genre classification [23, 26, 24, 19, 11, 36], emotion recognition [51, 29, 34], mood tagging [25], and instrument tagging [4, 45]. These tasks aim to build systems that automatically classify the semantics of music, which facilitate the automatic organization, retrieval, and recommendation of large-scale music databases [26]. In the past decade, deep learning approaches for these tasks have been proposed [26, 21, 10, 19, 37]. By training Deep Neural Networks (DNNs) to perform classification on acoustic features extracted from the original audio mixture, previous work has achieved promising performance on multiple datasets [21, 10].

However, current music classification models still struggle to perform music classification when there are many classes [14], possibly due to the intrinsic complexity of the music signal. In most of the music signals, multiple instruments are present in the performance and overlap with each other. Each instrument plays its own part and contributes to the overall semantics of the music. For example, the presence of a heavily distorted electric guitar serves as a strong discriminative indicator that a track belongs to the rock or metal genre, just as the noticeable electric bassline is highly characteristic of jazz-funk music. However, such subtle yet useful information may be obscured by other simultaneously performed instruments and vocals, potentially limiting the performance of music classification models.

Recently, deep learning-based Music Source Separation (MSS) models have achieved significant progress [13]. Models such as HT-Demucs [32] and Mel-Roformer [42] have achieved high Signal-to-Distortion Ratios (SDR) in the separation of the mixture into four stems: vocals, drums, bass, and others. Recent works such as Banquet [46] can even perform MSS on any instrument in a query-based setting. These recent advancements open up a possible way toward improving music classification performance: separate the mixture into several stems,

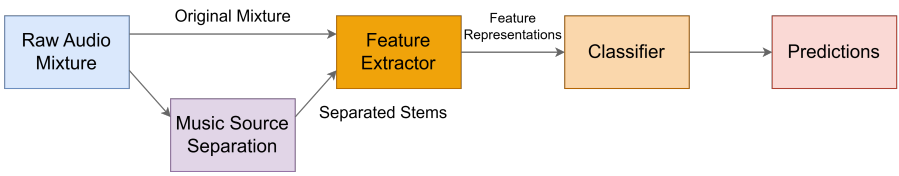


Figure 1: An overview of the proposed pipeline.

and then perform music classification on them. Through MSS, the downstream classification model can better access the features of individual instruments, potentially achieving better performance when dealing with complex mixture signals.

In this paper, we investigate the feasibility of using this separation-classification pipeline for music classification tasks. As shown in Figure 1, the proposed pipeline consists of three main components, namely *MSS*, *feature extractor*, and *classifier*. Given an input mixture audio, we first apply an MSS model to separate the mixture audio into several stems. Then, for each stem and the original mixture, we apply the feature extractor to extract high-level features from it. Finally, the classifier fuses all the features and performs music classification. Among all three components, we freeze the MSS model's and the feature extractor's pretrained weights and only train the classifier in the downstream tasks.

To evaluate the proposed pipeline, we conduct a set of experiments that involve three music classification tasks (genre classification, mood/theme tagging, and emotion recognition) and seven datasets (FMA-medium [8], GTZAN [41], ISMIR 2004 genre dataset [5], MTG-Jamendo [4], PMEMO [50], EmoMusic [35], and DEAM [1]). Experiment results suggest that combining the mixture and the individual stems extracted by an MSS model consistently improves the downstream music classification performance in genre classification. For emotion recognition and mood/theme tagging, the results are mixed and more dataset-dependent. We suspect that since mood is a more holistic attribute that relies on the interplay between instruments, the stem features are less effective. On the other hand, many music genres are famous for their instrumentation or the style of specific instruments, thus utilizing separated stems for music genre prediction is more beneficial. For reproducibility, we will release the source code and trained

models after the publication of this paper.

The main contributions of this paper are as follows:

- **Empirical study on the effectiveness of MSS and stem features for music classification.** We conduct the first systematic study of modern MSS models as a preprocessing step for music classification across three tasks and seven datasets with categorical (including single-label and multi-label) and dimensional annotation schemes.
- **MSS paradigm comparison.** We compare a fixed-stem model (HT-Demucs) and a query-based model (Banquet) in terms of downstream classification performance, finding that high-fidelity fixed-stem separation provides a better improvement for the classification task.
- **Task-type principle.** We identify a consistent principle: stem fusion benefits tasks where discriminative cues are instrument-localized (genre), but does not (emotion) or only marginally (mood/theme) improve performance on tasks that rely on the semantics of the entire mixture audio. This provides actionable guidance for future MIR system design.

2 Related Work

2.1 Music Source Separation

Music Source Separation (MSS), the task of separating mixture audio into multiple stems based on the instruments, has been a fundamental challenge in acoustic signal processing for more than 30 years [2]. Early approaches relied heavily on unsupervised statistical methods such as Non-negative Matrix Factorization (NMF) [17]. This approach of NMF led to the development of NMF-based MSS model that is specialized for rock music [49], which operates by minimizing the divergence between the target mixture spectrogram and the estimated source matrices. The use of NMF then continued into the implementation of Flexible Audio Source Separation Toolbox (FASST) model [33], which combined spectral factorization with spatial covariance models using expectation-maximization algorithms. Recently, the use of deep

learning models, such as U-Net for MSS, has significantly improved the separation performance [39, 40, 13, 16]. These MSS models can be categorized into three types based on the domain that they operate in: 1) **time-domain** MSS such as Wave-U-Net [39], Demucs [9], and Conv-TasNet [22], which directly predict the waveform for each stem; 2) **frequency-domain** MSS such as Spleeter [13] and Open-Unmix [40], which predict the spectrogram for each stem and then apply inverse short-time Fourier transform to produce the waveform; and 3) **hybrid** methods that operate in both frequency-domain and time-domain, such as HT Demucs [32] and Mel-RoFormer [42], which utilize the attention mechanisms and band-split modeling to capture long-range dependencies.

Traditionally, an MSS model separates the mixture into several predefined stems, such as *vocals*, *drums*, *bass*, and *others* (VDBO) defined in the MusDB18 dataset [30], or the 6-stem setting (VDBO+*piano* and *guitar*) used in the `htdemucs_6s` version of the HT Demucs model [32]. Although these definitions simplify the MSS task and allow for better comparability of results [28], they also introduce a limitation to MSS systems built on such problem definitions, as some instruments are not well-separated. For example, acoustic guitar and electric guitar are split into the same *guitar* stem. To address this issue, there have recently been MSS works that propose a highly flexible query-based problem definition [20], such as Banquet [46]. These models are trained to perform MSS based on an additional *query* audio, which specifies the target instrument to be isolated. Technically, these query-based MSS models can extract arbitrary instruments from the mixture. However, in practical applications, these query-based systems still lag behind conventional MSS models on standard instruments. In this work, we adopt a traditional 6-stem MSS model (HT Demucs [32]) and a query-based MSS model (Banquet [46]) in our experiments.

2.2 Using Stem-Level Features for MIR

Stem-level features have been utilized for many MIR applications, such as automatic music transcription [3, 48], singing-to-MIDI transcription [43], lyrics alignment [38], automatic lyrics transcription [44], and spatial audio upmixing [13]. Through MSS, the downstream task can focus on the specific stem of interest, such as the vocal stem for lyrics

transcription, significantly reducing the difficulty of these tasks.

In the context of music classification, the effectiveness of MSS is less pronounced, as music classification tasks do not explicitly focus on any single stem. Early work by Rosner *et al.* [31] proposed to separate the music signal into the harmonic and percussive components and then perform genre classification using a Support Vector Machine (SVM). However, their work was only evaluated in a dataset with 465 tracks, and their model relies on traditional hand-crafted features. Moreover, their method only separates music into the harmonic and percussive components, which is different from music source separation. The prior study closest to ours is EmoMucs [7], which utilizes a source separation model for music valence and arousal regression (based on Russell's 4Q model). Their experiment results show that the combination of features from different stems improves the performance of the valence regression, but not on the arousal regression. In this work, we focus on music classification instead of regression.

3 Methodology

3.1 System Overview

The primary objective of this study is to leverage MSS models to enhance the performance in music classification tasks. An overview of the proposed pipeline can be found in Figure 1, and a detailed version can be found in Figure 2. The proposed pipeline consists of three main components: a *MSS* model, a *feature extractor*, and a *classifier*. First, we apply an MSS model to extract the music signals of each stem. Then, for each stem and also the original mixture, the feature extractor is applied to extract the features from the audio. Finally, the classifier takes all the features as input and performs music classification.

In this section, we discuss each component of the pipeline one by one along with the models we choose for each component.

3.2 Music Source Separation (MSS)

As mentioned in Section 2.1, there are two MSS paradigms. The traditional paradigm separates music into pre-defined stems [32], while the query-based MSS paradigm extracts the instrument sound based

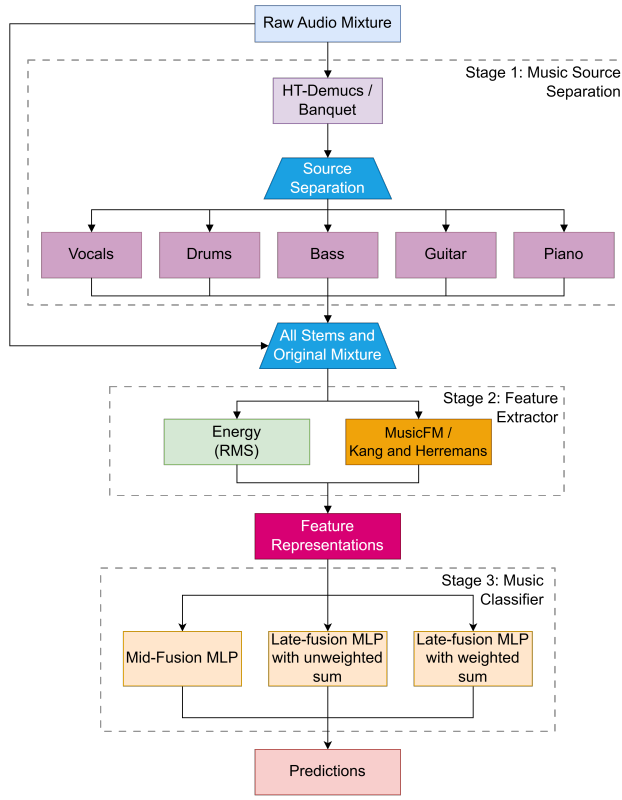


Figure 2: A detailed overview of the proposed methods.

on a query audio [20, 46]. In this work, we adopt one model for each paradigm. For traditional MSS, we adopt the 6-stem HT-Demucs model (`htdemucs_6s`), which separates the mixture into six stems: *vocals*, *drums*, *bass*, *piano*, *guitar*, and *others*. For query-based MSS, we adopt Banquet [46] as the MSS model.

3.3 Feature Extraction

For feature extraction, we employ three different feature extractors, namely MusicFM [47], RMS energy (loudness), and the feature extractor proposed by Kang and Herremans (hereinafter referred to as *Kang and Herremans*) [14].

MusicFM. MusicFM [47] is a Conformer model [12] pretrained

using BERT-RQ [6], a self-supervised learning (SSL) algorithm. With such an SSL pretraining, MusicFM is capable of extracting rich high-level features for music. We use the `MusicFM-MSD` checkpoint as the feature extractor and extract the framewise features after the seventh Conformer layer.[1] Then, we perform average pooling on the framewise features to obtain a feature representation for the entire audio.

RMS energy (loudness). In addition to MusicFM, we also compute the root mean square (RMS) energy of the entire audio, which reflects the overall loudness of the audio. When integrated into the proposed pipeline, the RMS feature of each stem serves as an indicator of the presence and/or importance of each instrument, which can be used by the downstream music classification model to make the prediction.

Kang and Herremans. Recently, Kang and Herremans [14] proposed a model for music classification, which takes the combination of MERT's features [18], key and chord as the input. Although the key and chord information are inferred by a key detection model and a chord recognition model, respectively, and are not guaranteed to be correct, the inclusion of these features still improve the music classification performance. Inspired by their success, we also employ the same feature extraction method in this work. We refer to their original paper for more details [14].

3.4 Music Classifier

For the downstream music classifier, we propose three systems, including mid-fusion Multi-Layer Perceptron (MLP), late-fusion MLP with unweighted sum, and late-fusion MLP with weighted sum. For the sake of description, we assume that there are N stems in total (which can include the original mixture) and the feature extractor extracts a d dimension feature vector for each stem. In practice, N and d could vary based on the choice of the MSS model (Section 3.2) and the feature extractor (Section 3.3).

Mid-fusion MLP. This is a three-layer MLP that takes the concatenation of all stems' features as the input and performs music classification. In other words, the input to this mid-fusion MLP is a $N \times d$ dimension vector.

[1] https://github.com/minzwon/musicfm

Table 1: Summary of the seven datasets used in this study. "V/A" denotes binary valence and arousal labels derived from continuous annotations.

Dataset	Task	Tracks	Labels	Label type
FMA-Medium [8]	Genre	25,000	16 genres	Multi-class
GTZAN [41]	Genre	1,000	10 genres	Multi-class
ISMIR04 [5]	Genre	1,458	6 genres	Multi-class
MTG-Jamendo [4]	Mood/theme tagging	17,982	56 tags	Multi-label
PMEmo [50]	Emotion	794	V / A	Binary
EmoMusic [35]	Emotion	744	V / A	Binary
DEAM [1]	Emotion	1,802	V / A	Binary

Late-fusion MLP with unweighted sum. This system consists of N three-layer MLPs, all with the same model architecture. Each MLP processes the features of one stem and performs music classification on them. Then, the predictions of each MLP are aggregated through an unweighted sum.

Late-fusion MLP with weighted sum. This system is similar to the "Late-fusion MLP with unweighted sum", utilizing the same N three-layer MLPs for initial stem classification. However, instead of using an unweighted sum for aggregation, it introduces a dynamic weighting mechanism that utilizes the stem's RMS feature to determine the weight for each stem. In particular, the RMS of each stem is fed into a dedicated two-layer MLP, which outputs a weight value between 0 and 1 via a sigmoid activation. Given these weight values, the predictions of each MLP (for each stem) are aggregated through weighted sum. This weighting mechanism allows the model to focus more on the active, acoustically dominant stems during classification, while paying less attention to silent stems.

4 Experiment Setup

4.1 Datasets

We evaluate the music classification systems on seven datasets that cover three task categories. Table 1 summarizes the key statistics; details follow below.

4.1.1 Genre Classification

FMA-Medium [8] contains 25,000 tracks (30s clips) spanning 16 unbalanced genre classes obtained from the Free Music Archive. The class distribution is highly skewed, with the most frequent genre outnumbering the least frequent by more than an order of magnitude. We use the official train/validation/test split provided with the dataset.

GTZAN [41] contains 1,000 tracks (30s clips) distributed evenly in 10 genres (100 tracks per genre). We use the 500/250/250 training/validation/test data split proposed in [15], which employs an artist filter to prevent artist leakage between splits. GTZAN's balanced class distribution makes it a controlled testbed for verifying that gains observed on FMA-Medium [8] are not artifacts of class imbalance.

ISMIR04 Genre Dataset (ISMIR04) [5] is a dataset collected for the Genre Identification task of the ISMIR 2004 audio description contest. It contains 1,458 full tracks (729 training, 729 test) across 6 genre classes: Classical, Electronic, Jazz/Blues, Metal/Punk, Rock/Pop, and World. Its genre categories are acoustically more distinct than those of FMA-Medium or GTZAN, providing a complementary evaluation point for genre classification.

4.1.2 Mood/theme Tagging

MTG-Jamendo [4] is a large-scale multi-label music tagging dataset providing multiple annotations. For mood/theme tagging, we use the *mood/theme* subset, which consists of 17,982 tracks annotated with 56 mood/theme tags. Each track may carry multiple simultaneous tags, and the distribution is severely imbalanced; many mood categories cover fewer than 2% of the tracks. We follow the official train/validation/test splits provided with the dataset for the experiment.

4.1.3 Emotion Recognition

For emotion recognition, we conduct evaluation on three datasets that provide binary valence and arousal labels, derived from continuous per-segment annotations, and follow the official splits provided by Kang and Herremans [14] for all three.

PMEmo [50] provides 794 clips with crowd-sourced per-segment

valence/arousal ratings collected via a mobile music player application.

EmoMusic [35] contains 744 clips (45s each) selected from the Free Music Archive to cover all four quadrants of Russell's valence-arousal model, providing balanced coverage of the affective space.

DEAM [1] (MediaEval Database for Emotional Analysis of Music) provides 1,802 audio clips with dynamic per-second valence and arousal annotations. We use the per-clip mean annotation as the label.

4.2 Implementation Details

MSS models. For HT Demucs [32], we use the `htdemucs_6s` checkpoint for MSS. For the query-based Banquet model [46], we follow HT Demucs using a 6-stem separation setting, but exclude the *others* stem, as it is not a straightforward "stem" in the context of query-based MSS. We use the audio samples obtained from SampleFocus[2] and Youtube[3] as query audios. Specifically, the audio queries include a duet voice of a male and female vocalist (vocals), a performance of drum kit (drums), a clean bass performance (bass), a classical piano performance (piano) and a clean electrical guitar performance (guitar).

Feature extractors. For MusicFM, we use the `MusicFM-MSD` checkpoint and extract representations from the 7th Conformer layer, following the sample usage code snippet in the official repo. For experiments using the Kang and Herremans feature extractor, we follow their original protocol: a frozen MERT-95M model [18] with features extracted from the 5th and 6th transformer layers. For the key and chord extraction, since the scripts from the original work were not publicly released, we re-implement the key and chord recognition models. Specifically, we use the pretrained Bi-directional Transformer for Chord Recognition (BTC) model [27] to obtain frame-level chord estimations. Then, we use the `mir_eval` and `pretty_midi` libraries to convert chords into MIDI note events, and use the `music21` toolkit to estimate the global key from the note events. This reconstruction may introduce minor deviations from the original pipeline, which we discuss when reporting results.

[2] https://samplefocus.com/
[3] https://www.youtube.com/

4.3 Evaluation Metrics

To evaluate the performance of the proposed pipeline, we tracked a standard suite of classification metrics: Accuracy, F1-Score, Area Under the Receiver Operating Characteristic Curve (ROC-AUC), and Area Under the Precision-Recall Curve (PR-AUC). While all metrics are reported to provide a complete performance profile, multi-label datasets like MTG-Jamendo suffer from severe class imbalance, where negative samples overwhelmingly outnumber positive samples. Consequently, accuracy plateaus near 100% since predicting all tags as absent already yields a high accuracy due to the "perfect prediction" of true negative samples, and ROC-AUC can remain uninformatively high because it measures ranking across all thresholds without penalizing poor precision on the minority positive class. The F1-score does not suffer from the class imbalance issue, but could be sensitive to the selection of a threshold that decides whether a tag/class is predicted. Therefore, **PR-AUC** serves as our primary evaluation criterion for these tasks, as it provides a highly sensitive measure of the model's ability to correctly identify rare subjective tags without being skewed by the dominant negative class.

4.4 Training Details

All models are trained using the Adam optimizer with a learning rate of 1×10^{-4}. For genre classification, we train the model using a batch size of 128 for 20 epochs and use the model checkpoint that maximizes the sum of accuracy, PR-AUC, and ROC-AUC in the validation set for evaluation. For mood/theme tagging and emotion recognition, we follow Kang and Herremans' training protocol [14]: batch size of 8, up to 500 epochs, with early stopping monitored on the validation score (patience of 10 epochs, minimum $\Delta = 10^{-4}$). For multi-label tasks that require a decision threshold to compute the F1-score and accuracy, we choose the threshold that maximizes the F1-score on the validation set for evaluation.

5 Results

5.1 Genre Classification

5.1.1 Results on FMA-Medium

In our first experiment, we evaluated genre classification performance on the FMA-Medium dataset [8] to answer two research questions: 1) whether incorporating source-separated instrument stems outperforms the baseline of utilizing only the unseparated mixture, and 2) if so, which MSS model (HT-Demucs or Banquet) leads to the best downstream performance. Since there are too many possible combinations (tasks, MSS models, the combination of stems, feature extractors, downstream classifiers), which leads to an inevitable combinatorial explosion, we are unable to examine all combinations. Therefore, we first focus on the choice of MSS models and stems. For the feature extractor and classifier, we adopt MusicFM and mid-fusion MLP, respectively.

Experiment results are shown in Table 2 (for Banquet) and 3 (for HT-Demucs). We have the following observations.

- Regardless of the choice of MSS models, utilizing only one stem always results in poorer performance. This observation suggests that none of the separated stems can replace the role of the original mixture for downstream genre classification. This is as expected. Since each instrument may carry its own musical semantics, only considering one instrument stem would result in a loss of information.
- When using the same combination of stems, using HT-Demucs for MSS (Table 3) generally performs better than using Banquet (Table 2), with only a few exceptions where both MSS models yield similar performance or Banquet performs better (e.g., drums only). For example, combining all stems extracted with Banquet leads to an accuracy, F1 score, ROC-AUC, PR-AUC of 0.6768, 0.4584, 0.8915, 0.5084, respectively; while combining all stems extracted with HT-Demucs leads to an accuracy, F1 score, ROC-AUC, PR-AUC of 0.6807, 0.4632, 0.8951, 0.5425, respectively.
- Generally, combining more separated stems yields better performance. Fusing features from all five stems extracted with HT-

Table 2: Genre classification results on FMA-Medium (Banquet-separated stems). All the settings here use Mid-fusion MLP as the classifier. Abbreviations: Acc = Accuracy, G = Guitar, D = Drum, P = Piano, B = Bass, V = Vocal.

Features	**Acc (%)**	**F1-score**	**ROC-AUC**	**PR-AUC**
Baseline				
Mixture Only	68.82	0.4695	**0.9069**	0.5150
Single Stem Features				
G Only	60.38	0.3423	0.8724	0.4102
D Only	64.62	0.3925	0.8917	0.4662
P Only	61.29	0.3682	0.8729	0.4152
B Only	60.11	0.3374	0.8696	0.4052
V Only	60.89	0.3687	0.8560	0.4049
Four-Stem Fusion (Leave One Out)				
G + D + P + B	66.32	0.4456	0.8932	0.5066
G + D + P + V	66.90	0.4373	0.8892	0.5097
G + D + B + V	67.11	0.4588	0.8861	0.5043
D + P + B + V	66.78	0.4392	0.8890	0.5030
All (Five)-Stem Fusion				
All Stems	67.68	0.4584	0.8915	0.5084
Mixture + Stem Fusion				
Mixture + G + D + P	68.54	0.4730	0.8944	0.5230
Mixture + G + D + P + B	68.73	0.4684	0.8970	0.5236
Mixture + All Stems	**68.89**	**0.4833**	0.9009	**0.5306**

Demucs leads to a performance similar to only using the mixture, in which *All Stems* outperforms *Mixture Only* in PR-AUC by around 2.75%, but underperforms *Mixture Only* in accuracy, F1 score, and ROC-AUC by 0.75%, 0.63%, and 1.18%, respectively. These results show that the fusion of stem features do improve the performance, but this method at most only performs roughly on par with the *Mixture Only* baseline.

- Combining the mixture audio with the stems separated with HT-Demucs leads to better performance than *Mixture Only*, as shown in Table 3. These results confirm the effectiveness of employing a MSS model for music genre classification. They also show the complementary property of mixture and separated stems in music genre classification.

Considering these experiment results, in the remainder of this pa-

Table 3: Genre classification results on FMA-Medium (HT-Demucs-separated stems). All the settings here use Mid-fusion MLP as the classifier. Abbreviations: Acc = Accuracy, G = Guitar, D = Drum, P = Piano, B = Bass, V = Vocal.

Features	**Acc (%)**	**F1-score**	**ROC-AUC**	**PR-AUC**
Baseline				
Mixture Only	68.82	0.4695	0.9069	0.5150
Single Stem Features				
G Only	62.35	0.3868	0.8710	0.4423
D Only	63.67	0.3862	0.8425	0.4478
P Only	61.26	0.3913	0.8687	0.4386
B Only	61.10	0.3608	0.8657	0.4188
V Only	61.69	0.3661	0.8602	0.4122
Four-Stem Fusion (Leave One Out)				
G + D + P + B	66.08	0.4374	0.8939	0.5074
G + D + P + V	68.84	0.4631	0.8868	0.5177
G + D + B + V	68.88	0.4821	0.8924	0.5393
D + P + B + V	67.25	0.4421	0.8852	0.5006
All (Five)-Stem Fusion				
All Stems	68.07	0.4632	0.8951	0.5425
Mixture + Stem Fusion				
Mixture + G + D + P	**70.05**	**0.4878**	0.8988	0.5372
Mixture + G + D + P + B	69.43	0.4744	0.9029	**0.5442**
Mixture + All Stems	68.81	0.4829	**0.9081**	0.5261

per, we only conduct experiments with HT-Demucs, as it generally leads to a better classification performance. For the selection of stems, we recognize that *Mixture+G+D+P*, *Mixture+G+D+P+B*, and *Mixture+All Stems* perform comparably well with each other. Therefore, we report the results with these three settings if possible.

Our next FMA-Medium experiments focus on the comparison of different music classifier. In particular, we compare **Mid-fusion MLP** with **Late-fusion with unweighted sum** and **Late-fusion with weighted sum**. As shown in Table 4, generally, all three methods perform comparably well with each other. None of the methods outperforms others in all three combinations of stem. These results show that the choice of the downstream music classifier does not significantly affect the music genre classification performance. Interestingly, almost all settings produce a better F1-score, and all settings produce a bet-

Table 4: Comparison of mid-fusion MLP, late-fusion MLP with unweighted sum, and late-fusion MLP with weighted sum on the FMA-Medium dataset (music genre classification).

Features	Accuracy (%)	F1-Score	ROC-AUC	PR-AUC
Baseline				
Mixture Only	68.82	0.4695	0.9069	0.5150
Mid-fusion MLP				
Mixture + G + D + P	**70.05**	**0.4878**	0.8988	0.5372
Mixture + G + D + P + B	69.43	0.4744	0.9029	**0.5442**
Mixture + All Stems	68.81	0.4829	**0.9081**	0.5261
Late-fusion with Unweighted Sum				
Mixture + G + D + P	68.57	**0.4892**	0.9008	0.5322
Mixture + G + D + P + B	**69.19**	0.4851	0.8994	**0.5373**
Mixture + All Stems	68.65	0.4861	**0.9040**	0.5294
Late-fusion with Weighted Sum				
Mixture + G + D + P	67.56	0.4479	0.8987	0.5233
Mixture + G + D + P + B	**69.16**	**0.4916**	**0.9082**	0.5340
Mixture + All Stems	68.81	0.4854	0.9070	**0.5349**

ter PR-AUC compared to the *Mixture Only* baseline, while performing comparably well or better than the baseline in accuracy and ROC-AUC. These results further demonstrate the effectiveness of including stem features to improve genre classification performance.

In summary, the FMA-Medium results establish that downstream classification improvements depend on the performance of the MSS model. With the use of a highly effective MSS model (HT-Demucs), all music classifiers lead to a better music genre classification performance compared to the mixture-only baseline.

5.1.2 Results on GTZAN

To validate the generalization of the proposed methods beyond the FMA-Medium dataset, a similar series of late-fusion experiments are conducted on the GTZAN dataset [41]. As detailed in Table 5, the integration of separated stem features along with the original mixture consistently improves performance in almost all configurations in all metrics. The best proposed method, **Late-fusion with unweighted sum** with *Mixture+All Stems*, achieves an improvement of 5.6% in

Table 5: Genre classification results on the GTZAN dataset.

Features	Accuracy (%)	F1-Score	ROC-AUC	PR-AUC
Baseline				
Mixture Only	84.40	0.8386	0.9811	0.9161
Late-fusion with Unweighted Sum				
Mixture + G + D + P	86.40	0.8669	0.9856	0.9329
Mixture + G + D + P + B	88.40	0.8822	0.9862	0.9391
Mixture + All Stems	**90.00**	**0.9010**	**0.9897**	**0.9507**
Late-fusion with Weighted Sum				
Mixture + G + D + P	85.20	0.8517	0.9815	0.9140
Mixture + G + D + P + B	88.00	0.8799	0.9838	0.9291
Mixture + All Stems	**88.80**	**0.8846**	**0.9878**	**0.9432**

Table 6: Genre classification results on the ISMIR04 dataset.

Features	Accuracy (%)	F1-Score	ROC-AUC	PR-AUC
Baseline				
Mixture Only	83.13	0.7323	0.9716	0.8448
Late-fusion with Unweighted Sum				
Mixture + G + D + P	88.20	0.8412	0.9810	0.9215
Mixture + G + D + P + B	87.24	0.8462	0.9810	0.9252
Mixture + All Stems	**89.17**	**0.8761**	**0.9838**	**0.9371**
Late-fusion with Weighted Sum				
Mixture + G + D + P	85.32	0.8049	0.9780	0.9038
Mixture + G + D + P + B	85.73	0.8053	0.9771	0.9063
Mixture + All Stems	**89.30**	**0.8648**	**0.9816**	**0.9274**

accuracy and 3.46% in PR-AUC. The inclusion of B (bass) and V (vocals) also consistently improves performance. Interestingly, the use of weighted sum slightly harms the overall performance.

5.1.3 Results on ISMIR04

Then, we evaluate the proposed methods on the ISMIR04 dataset [5]. As detailed in Table 6, the findings are consistent with previous experiments, confirming that multi-feature fusion outperforms the mixture-only baseline. By combining the features of the mixture and all five source-separated stems, both methods improve the classification per-

Table 7: Mood/Theme tagging results on the MTG-Jamendo dataset (mood/theme subset). All methods here use the MusicFM features as input.

Features	Accuracy (%)	F1-Score	ROC-AUC	PR-AUC
Baseline				
Mixture Only	92.70	0.0771	0.7381	0.0781
Late-fusion with Unweighted Sum				
Mixture + G + D + P	92.23	0.0775	0.7413	**0.0790**
Mixture + G + D + P + B	**92.27**	0.0778	**0.7432**	0.0777
Mixture + All Stems	92.19	**0.0804**	0.7374	0.0781
Late-fusion with Weighted Sum				
Mixture + G + D + P	92.40	**0.0882**	0.7402	0.0804
Mixture + G + D + P + B	**92.73**	0.0820	**0.7441**	0.0798
Mixture + All Stems	92.38	0.0872	0.7429	**0.0812**

formance by more than 6% accuracy, 13% F1-score, and 8% PR-AUC score. These improvements show that the effectiveness of including source-separated stems for music genre classification can be observed across multiple datasets.

5.2 Mood/Theme Tagging

To extend the proposed methods beyond genre classification, we then evaluate them on the MTG-Jamendo dataset's mood/theme subset [4]. We first use MusicFM as the feature extractor. As detailed in Table 7, the results are mixed. The integration of stem features generally improves F1-score, ROC-AUC and PR-AUC slightly, but reduces the accuracy slightly. All improvements are within the margin of 1%. These results differ from those of genre classification, in which the improvements are clearer.

To further investigate whether the stem features improve mood/theme tagging performance, we replace the MusicFM feature extractor with Kang and Herremans [14], where MERT features and automatically predicted chords and key are combined and fed into the classifier. For this experiment, we adopt mid-fusion as the music classifier, which aligns better with Kang and Herremans' original setting [14]. As detailed in Table 8, our reproduction of the official mixture-only baseline yields a ROC-AUC of 0.7667 and a PR-AUC of 0.1279, below the paper's 0.7806 and 0.1521. We attribute this gap to two factors: 1) the

Table 8: Mood/Theme tagging results on the MTG-Jamendo dataset (mood/theme subset). All methods here use the Kang and Herremans features (MERT features+chords+key) as input.

Features	**ROC-AUC**	**PR-AUC**
Baselines		
Mixture Only (Paper Result)	**0.7806**	**0.1521**
Mixture Only (Our Best Reproduced Result)	0.7667	0.1279
Mid-fusion MLP		
Mixture + All Stems	0.7768	0.1421

Table 9: Valence classification results on the PMEmo dataset. All methods here use the Kang and Herremans features (MERT features+chords+key) as input, and use the mid-fusion MLP as the classifier.

Features	**Accuracy (%)**	**F1-Score**	**ROC-AUC**	**PR-AUC**
Mixture Only	81.16	0.8693	**0.8731**	**0.9459**
Mixture + All Stems	**82.61**	**0.8775**	0.8721	0.9446

key and chord extraction scripts were unavailable, necessitating the reconstruction pipeline described in Section 4.2; and 2) the hyperparameters of the classifier differ from those of the original paper. We have tried several settings to reproduce the results of the original paper by modifying the classifier architecture and training hyperparameters to align with the original paper's setting. However, all attempts did not yield the results reported in their paper. Therefore, we eventually stick to the proposed mid-fusion setting. In this setting, the use of separated stems improves ROC-AUC and PR-AUC by 1.01% and 1.42%, respectively. These results, combined with the late fusion results in Table 7, suggest that the use of stem features improves the mood/theme tagging performance, but only by a small margin compared to that of genre classification.

5.3 Emotion Classification

The experiment results in genre classification and mood/theme tagging suggest that the effectiveness of including source-separated stem features in music classification varies across tasks. We suspect that the characteristics of the tasks play an important role. While some genres

Table 10: Arousal classification results on the PMEmo dataset. All methods here use the Kang and Herremans features (MERT features+chords+key) as input, and use the mid-fusion MLP as the classifier.

Features	Accuracy (%)	F1-Score	ROC-AUC	PR-AUC
Mixture Only	84.93	0.8976	**0.9219**	0.9724
Mixture + All Stems	**85.22**	**0.9009**	0.9210	**0.9725**

are closely linked to the style of a particular instrument (e.g., heavy metal is associated with heavily distorted electric guitars), emotion-related tags are less closely linked to individual instruments, but more related to the overall semantics of the entire music. As a result, emotion-related tasks benefit less from the inclusion of stem features.

To validate this assumption, we further conduct experiments in music emotion recognition. In particular, we adopt the *binary classification* formulation of emotion recognition following Russell's 4Q model, by considering arousal and valence classification individually. We apply the same Kang and Herremans feature extractor to three emotion recognition datasets: PMEmo [50], EmoMusic [35], and DEAM [1].

5.3.1 Results on PMEmo

As detailed in Tables 9 and 10, the results on PMEmo are mixed. For valence classification, *Mixture+All Stems* outperforms *Mixture Only* by 1.45% in accuracy and 0.82% in F1-score, but underperforms by 0.1% and 0.13% in ROC-AUC and PR-AUC score, respectively. For arousal classification, *Mixture+All Stems* slightly outperforms *Mixture Only* by 0.29%, 0.33%, and 0.01% in accuracy, F1-score, and PR-AUC, respectively, while slightly underperforms *Mixture Only* by 0.09% in ROC-AUC. Similar to the results in MTG-Jamendo's mood/theme tagging, most of these performance differences are within 1%, suggesting that the use of source-separated stem features does not significantly improve the emotion classification performance.

5.3.2 Results on EmoMusic

On the EmoMusic dataset, as shown in Tables 11 and 12, the results are again split across metrics. For valence, the mixture-only baseline

Table 11: Valence classification results on the EmoMusic dataset. All methods here use the Kang and Herremans features (MERT features+chords+key) as input, and use the mid-fusion MLP as the classifier.

Features	**Accuracy (%)**	**F1-Score**	**ROC-AUC**	**PR-AUC**
Mixture Only	**77.46**	0.7455	0.8132	0.8631
Mixture + All Stems	72.80	**0.7621**	**0.8231**	**0.8723**

Table 12: Arousal classification results on the EmoMusic dataset. All methods here use the Kang and Herremans features (MERT features+chords+key) as input, and use the mid-fusion MLP as the classifier.

Features	**Accuracy (%)**	**F1-Score**	**ROC-AUC**	**PR-AUC**
Mixture Only	83.73	0.8338	**0.9387**	**0.9366**
Mixture + All Stems	**84.27**	**0.8393**	0.9344	0.9355

achieves higher raw Accuracy (77.46% vs. 72.80%), while the mid-fusion *Mixture+All Stems* model achieves a higher F1-Score (0.7621 vs. 0.7455), ROC-AUC (0.8231 vs. 0.8132), and PR-AUC (0.8723 vs. 0.8631). For arousal, the *Mixture+All Stems* model improves accuracy (84.27% vs. 83.73%) and F1-Score (0.8393 vs. 0.8338), while the *Mixture Only* model retains a small edge in ROC-AUC (0.9387 vs. 0.9344) and PR-AUC (0.9366 vs. 0.9355). While the performance differences are not as modest as those in the PMEmo, again, none of the models achieves a convincing performance gain against each other.

5.3.3 Results on DEAM

The final set of emotion classification experiments was conducted on the DEAM dataset, with the results shown in Tables 13 and 14. Surprisingly, the *Mixture Only* setting outperforms the *Mixture + All Stems* model in every metric in both valence and arousal classification. Noticeably, the performance gap in valence classification (Table 13) is quite clear, where *Mixture Only* outperforms *Mixture + All Stems* by 2.46% accuracy, 3.94% F1-score, 2.8% ROC-AUC score, and 2.94% PR-AUC. These results indicate that the use of source-separated stem features does not help emotion classification. In fact, performance degradation indicates that the classifier may overfit the stem features that are

Table 13: Valence classification results on the DEAM dataset. All methods here use the Kang and Herremans features (MERT features+chords+key) as input, and use the mid-fusion MLP as the classifier.

Features	Accuracy (%)	F1-Score	ROC-AUC	PR-AUC
Mixture Only	**79.38**	**0.7764**	**0.8800**	**0.8361**
Mixture + All Stems	76.92	0.7370	0.8520	0.8067

Table 14: Arousal classification results on the DEAM dataset. All methods here use the Kang and Herremans features (MERT features+chords+key) as input, and use the mid-fusion MLP as the classifier.

Features	Accuracy (%)	F1-Score	ROC-AUC	PR-AUC
Mixture Only	**80.62**	**0.7875**	**0.8957**	**0.8783**
Mixture + All Stems	80.37	0.7731	0.8909	0.8642

actually non-informative, resulting in suboptimal performance.

To sum up, from the results on the three emotion classification datasets, we can conclude that the use of separated stems' features does not clearly improve the emotion classification performance, and may even negatively affect the classification performance in some cases.

5.4 Discussion

Experiment results for three tasks reveal a discrepancy in the performance gain from the use of source-separated stem features between tasks. For genre classification, performance gains are generally noticeable. Among three genre classification datasets, the FMA-Medium dataset benefits the least from the use of stem features. Nevertheless, Table 4 still shows an improvement of around 2% in F1-score and around 2∼3% in PR-AUC score in multiple settings on FMA-Medium. For mood/theme tagging, the performance gain is less prominent. Even the best setting (mid-fusion MLP with *Mixture+All Stems*) only improves the PR-AUC by 1.42%. There are also many settings reported in Table 7 that only marginally improve performance. Finally, for emotion classification, the performance improvement of using stem features is insignificant on the PMEmo dataset (Tables 9 and 10) and the EmoMusic dataset (Tables 11 and 12), and is even negative on the DEAM

dataset (Tables 13 and 14).

In Section 1, we suspected that this discrepancy comes from whether the discriminative cues for the task are instrument-localized or not. This seems to explain why genre classification benefits more from stem features (as many genres are famous for the style/presence of certain instruments), while emotion classification does not. Regarding MTG-Jamendo's mood/theme tagging, after examining the list of tags,[4] we found that the mood/theme tagging task is somewhat between genre classification and emotion classification. In mood/theme tagging, there are tags more related to emotion, such as `calm`, `happy`, and `sad`. However, there are also tags more related to music genre or music style, such as `ballad`, `retro`, and `relaxing`,[5] As a result, the classification of some tags benefit more from the use of stem features, while others do not, resulting in a marginal improvement in the overall classification performance.

6 Conclusion

In this work, we propose the use of automatically separated stems to enhance music classification performance. By fusing the features from the original mixture and the separated stems, the downstream music classification model can have better access to per-stem details that may otherwise be obscured by the sound of other stems. Through experiments, we found that this method works well in genre classification, but less well in emotion classification, possibly because genres are usually associated with the style or presence of specific instruments, while emotions are more related to the holistic semantics of the entire mixture. These observations provide insights on the applicability of source-separated stems in downstream music classification tasks.

In addition to the contributions, we also recognize two limitations of this work. First, due to resource constraints, we were unable to sweep all combinations of music source separation models, feature extractors, and downstream music classifier architectures. Second, we were unable to reproduce some of the state-of-the-art music classification models

[4] https://github.com/MTG/mtg-jamendo-dataset/blob/master/data/tags/moodtheme.txt

[5] We believe these tags belong to the "theme" category of tags.

in our experiments, and thus our experiments are more focused on the comparison of the same model under different feature sets (e.g., mixture-only vs. mixture + separated stems). Future work can be conducted to mitigate these issues.

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